# Scoring chiplet interconnect simulations against exact references: ghost replicas, bond qubits and patch seams for a multi-chip Nighthawk

ThereminQ project (twobombs)
https://github.com/twobombs/thereminq-examples

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## Abstract

A six-chiplet, 720-qubit device of the IBM Nighthawk family can only be simulated approximately, and the usual mirror-circuit observable cannot tell a good chip-to-chip approximation from a bad one. We scale the problem down to fields of at most 24 qubits, where the whole register is also simulated exactly, and score 142 configurations against that reference. Between chiplets, replicas of the link partners (the *ghost loom*) hold a classical fidelity near 0.75 for three links, independent of depth from 6 to 12 cycles; mean-field and sampled couplings are indistinguishable from omitting the links (0.49); a bond-qubit bridge (QrackBridge) is exact until its bonds saturate and needs at least as many bond qubits as links. Inside a chiplet, patch seams (QrackAceBackend) keep an 8-qubit block fidelity of 0.95 to 0.98 at all depths, whereas a bridge cut with fewer bonds than couplers falls below that by depth 8. Link position, disabled-qubit position and chip topology do not change the ghost-loom result. Within six 8 GB accelerators the recommended full-size configuration is ACE patches inside, ghost loom between, ten links per pair.

## 1 Introduction

IBM's Nighthawk processors couple 120 qubits on a square lattice and are planned to be joined by l-couplers into multi-chip systems [1]. We model a third revision ("r3") as six 120-qubit chiplets on a $2 \times 3$ grid with ten l-couplers per facing pair; the layout is our assumption, not a published specification. A statevector of 720 qubits is out of reach, so the simulation is assembled from approximations at two levels: inside a chiplet, and between chiplets. Each level offers several engines from the Qrack ecosystem [2, 3].

The observable used at scale is the mirror Hamming signal $1 - 2h$, with $h$ the mean Hamming weight per qubit after $UU^\dagger$. It needs no reference and equals 1 for an exact simulator. It measures reversibility, however, and a simulator that drops every inter-chip gate is perfectly reversible. Choosing between interconnect methods therefore needs a ground truth. This report obtains one by scaling down.

## 2 Methods

**Fields.** All fields use the lattice, four-colour coupler schedule and circuit generator of `nighthawk_qrack.py`: each cycle applies Haar-random single-qubit gates to every qubit and CZ on one coupler colour. Links take a colour that is free at both ends, so each link fires once every four cycles. Three field types are used (Fig. 1): two chips of $3 \times 4$ qubits with $K = 1$ to 3 links (24 qubits); one chip of $4 \times 5$ qubits (20 qubits); six chips of $2 \times 2$ qubits on a $2 \times 3$ grid with one link per pair (24 qubits). A variant of the two-chip field has $3 \times 5$ cells with three disabled qubits, which frees a coupler colour at the chip centre.

**Engines.** *Exact*: one QrackSimulator. *ACE*: QrackAceBackend, which splits a register into patches joined at seam qubits; registers are written size:columns:rows. *Bridge*: QrackBridge, patches joined by at most $b$ bond qubits per cut, truncated beyond that. *Loom*: each chip runs in its own simulator and an outer layer carries the links. In *ghost* mode a chip holds a replica of each link partner and applies the link CZ exactly on its own copy; in *mean* mode the link becomes a phase $\pi P(1)$ of the partner; *sample* draws that phase from a sampled bit. *Cut* omits the links. A *stitched* readout of the ghost loom conditions one chip's sample on the other's ghosts.

**Scores.** With $p$ the exact output distribution of $U|0\rangle$ and $q$ the configuration's, the primary score is the classical fidelity $F_c = \left(\sum_x \sqrt{p_x q_x}\right)^2$. For looms $q$ is the product of the chips' distributions; for the bridge it is read from its state. ACE exposes no state, so every configuration is also scored by $F_8$, the same quantity on 8-qubit blocks inside a chip or around a link, estimated from 8192 samples for ACE. Linear XEB and the mirror signal are recorded for comparison. Depths are 4, 6, 8 and 12 cycles with 3 to 4 circuit instances; error bars are standard errors over instances. In total 142 configurations and 1500 scored points were run on two CPU cores.

## 3 Results

### 3.1 Between chiplets

Figure 2 shows the interconnect methods with exact chips. For three links the ghost loom gives $F_c = 0.75$ at depth 8 and 0.74 at depth 12. The mean-field loom

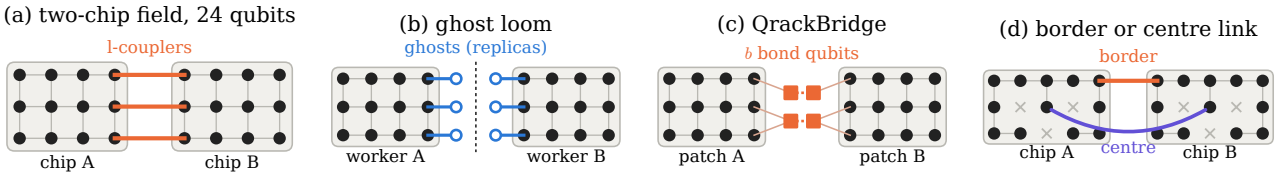



**Figure 1:** Schematics. (a) Two chips joined by l-couplers; the exact reference simulates all 24 qubits. (b) Ghost loom: each chip is simulated alone with replicas of its link partners. (c) QrackBridge: the cut is carried by $b$ bond qubits. (d) Field with three disabled qubits per chip (crosses), allowing a link at the chip centre as well as at the border.

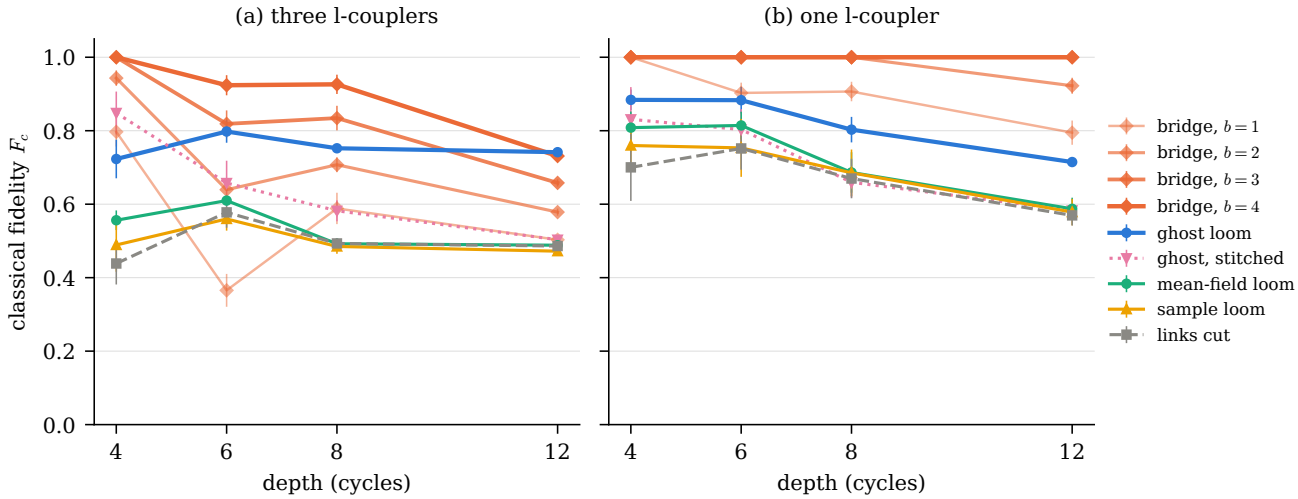



**Figure 2:** Classical fidelity of the output distribution against depth for two exact $3 \times 4$ chips. The interconnect is the only approximation.

(0.49), the sample loom (0.49, 0.47) and the cut baseline (0.49) coincide: phases derived from marginals add nothing that survives in the output distribution. The bridge is exact while the bonds hold the entanglement across the cut and then decays with depth; Table 1 gives the values. With $b = 1$ for three links it is below the cut baseline at depth 6 (0.37 against 0.58). At depth 8 it overtakes the ghost loom at $b = K$, at depth 12 only at $b > K + 1$ (Fig. 3a). The stitched readout improves the ghost loom at depth 4 (0.85 against 0.72) and degrades it from depth 6 (0.58 against 0.75 at depth 8). An ACE seam used as the interconnect gives $F_8 = 0.95$ across the link at depth 8, against 0.985 for the ghost loom.

Every additional link lowers the fidelity of every approximate method (Fig. 3b). Figure 3c plots the mirror signal against $F_c$: the cut baseline sits at signal 1.00 for any fidelity, and the mean-field loom at 0.27 for the same $F_c = 0.49$.

## 3.2 Inside a chiplet

On the $4 \times 5$ chip a row cut crosses five couplers. The bridge with two patches reaches $F_c = 0.25$, 0.39, 0.52 and 0.75 at depth 8 for $b = 2$, 3, 4 and 6 (Fig. 4a). On blocks, ACE registers hold $F_8 = 0.95$ to 0.98 at every depth, with the error present from the first cycle; the bridge with $b = 3$ is ahead at depth 6 (0.98) and behind at depth 8 (0.85) and 12 (0.81), and matches ACE only at $b = 6$ (Fig. 4b). XEB exceeds 1 for the truncated bridge (up to 1.7), so it was not used for ranking.

Table 2 combines both levels. ACE inside the chips is the larger loss ($F_8$ inside from 0.99 to about 0.92); the ghost loom keeps a lead of about 0.05 across the link over mean-field and cut. Where three bonds cover the three couplers of the cut, a bridge inside the chips is ahead of ACE; with two bonds they are level. A single bridge chain over both chips reaches 0.99 at $b = 3$ and 0.83 at $b = 1$. A ghost needs a free ACE site coupled to its anchor: a 12-site register for a 12-qubit chip has none, a 16-site register room for two of three.

## 3.3 Placement, disabled qubits, topology

For the ghost loom the row of the border makes no difference (0.80 corner row, 0.79 centre row), nor does linking chip centres instead of border corners (0.86, 0.87), although a centre link matters more to the device: omitting it costs more (cut 0.69 against 0.79). Placing the disabled qubits next to the link anchors gives no gain (0.74 against 0.75) and lowers the mean-field loom (0.54 against 0.60) (Fig. 5a,b).

With six chips the ghost loom gives 0.57 to 0.58 on

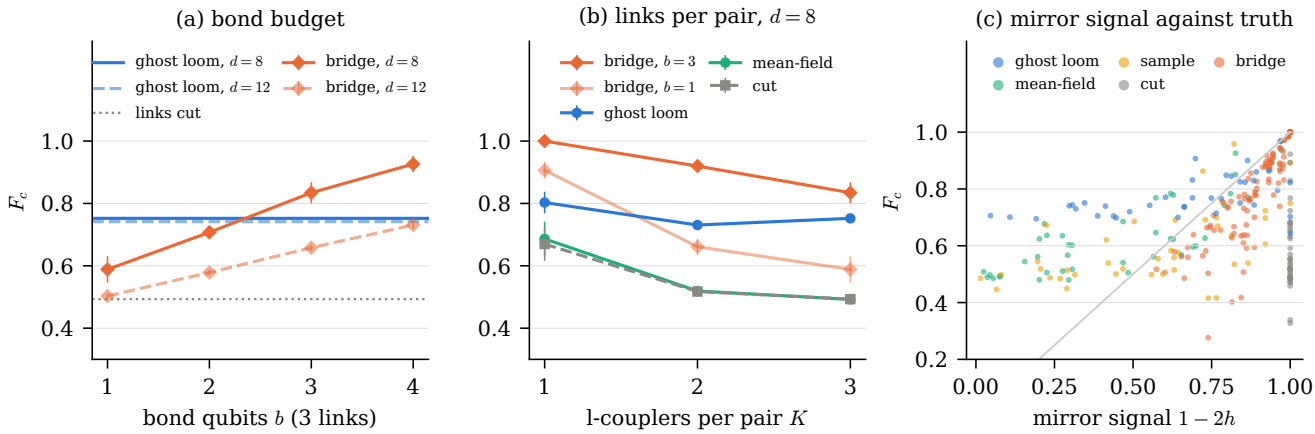



**Figure 3:** (a) Bridge fidelity against bond qubits for three links, with the ghost-loom level. (b) Fidelity against links per pair at depth 8. (c) Mirror signal against classical fidelity for every scored point of the two-chip fields.

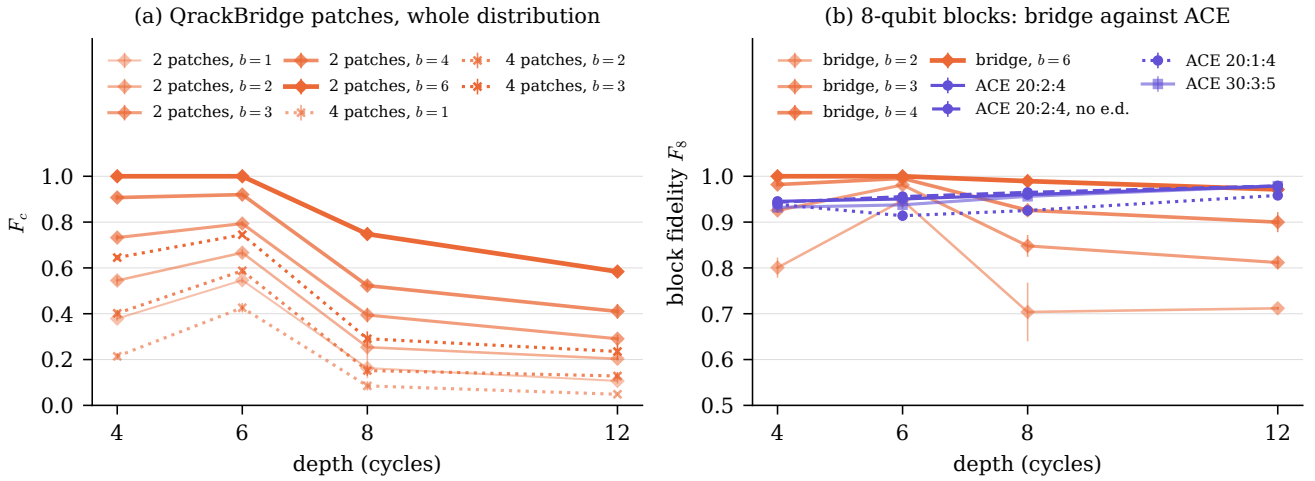



**Figure 4:** One $4 \times 5$ chip without interconnect. (a) QrackBridge with two or four row patches. (b) Block fidelity of the bridge against ACE registers (e.d.: error detection).

the grid, the ring and the grid with a long link, against 0.25 to 0.30 for mean-field and 0.18 for cut. A bridge chain with $b = 2$ reaches 0.39 on the grid; a bridge ring is exact on the ring it matches and 0.71 on the grid (Fig. 5c). The one-dimensional bridge is therefore tied to the chip topology in a way the loom is not.

### 3.4 Full-size resources

Table 3 lists the ghost-loom plan for the full device with the 120-qubit ACE preset, as computed by `field plan`. Ten links per pair fit six 8 GB accelerators with 64 of 72 links receiving a ghost. A bridge would need about ten bond qubits per cut at this link count, and its patches would have to hold 120-qubit chips exactly; neither fits.

## 4 Discussion

The ghost loom's plateau and the bridge's decay have the same origin seen from two sides. A replica captures the first-order effect of a link on each chip exactly and discards cross-chip correlations at readout, so its error is fixed by the product form and does not accumulate. A bond-limited cut is exact until the entanglement exceeds $2^b$ and then loses weight at every link layer. Which one wins is set by $b/K$ and depth; at ten links per pair the bridge is not available within the memory budget, and the ghost loom is.

Inside a chiplet the same ratio decides between bridge and ACE. A full-size chiplet cut crosses ten couplers, for which the affordable bond count is about six by our estimate, so the ratio is nearer the $4 \times 5$ test (bridge behind ACE by depth 8) than the two-chip test (bridge ahead).

**Recommendation.** ACE patches inside each chiplet; ghost loom between chiplets; ten links per pair on the default memory limits; border anchors; spare ACE sites coupled to the anchors rather than disabled qubits at the interconnect; the plain grid. The mean-field and sample looms should be treated as cut baselines.

**Limitations.** Chips of 4 to 20 qubits saturate entanglement quickly, and seam qubits are a larger fraction than at 120 qubits; the results are trends and mechanisms, not predictions of r3 fidelities. ACE is scored on blocks only. Three to four instances per point give standard errors of 0.01 to 0.05. The ghost loom at full size was planned but not run. The mirror signal remains valid for comparing chips under one interconnect;

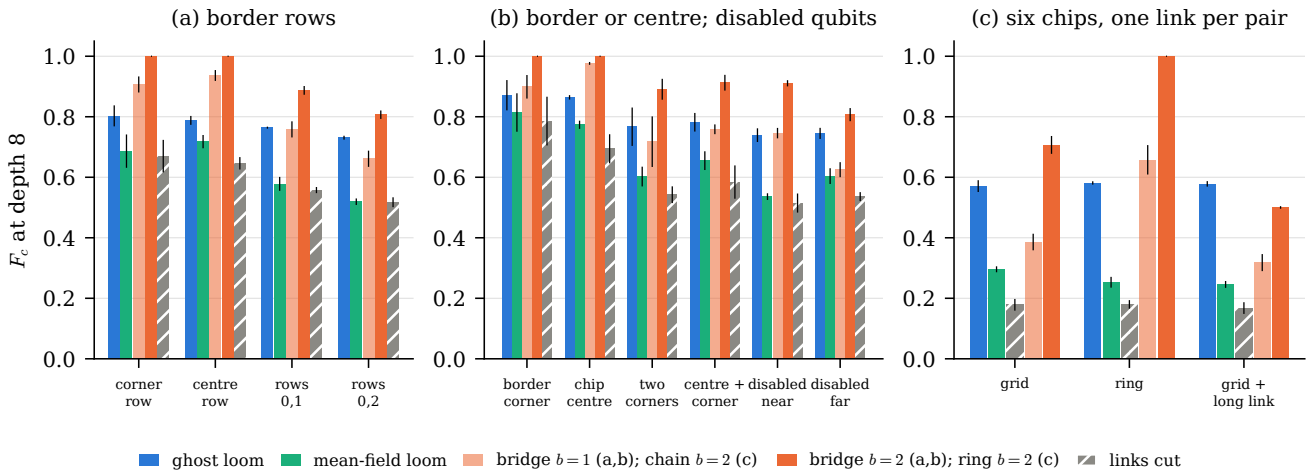



**Figure 5:** Classical fidelity at depth 8 with exact chips. (a) Rows of the facing border used by one or two links. (b) Border against chip-centre links, and three disabled qubits next to or far from the link anchors. (c) Six $2 \times 2$ chips on a grid, a ring, and a grid with one long link.

it cannot compare interconnects.

## Reproducibility

`nighthawk_qrack.py lab -exp inter mixed links inside rows holes numa` produces the records; `lab -selftest` checks that the exact engine and a bridge with spare bonds score 1; `nighthawk_graph.py -lab ... -lab-poster` draws the overview posters. PyQrack 2.30.6, Qrack 10.14.2, CPU only. The lab harness, runs and figures were produced with Claude (Anthropic).

## References


[1] IBM, "IBM delivers new quantum processors, software, and algorithm breakthroughs on path to advantage and fault tolerance," press release, 12 Nov. 2025.

[2] D. Strano *et al.*, "Exact and approximate simulation of large quantum circuits on a single GPU," arXiv:2304.14969 (2023).

[3] PyQrack, including QrackAceBackend and QrackBridge, https://github.com/unitaryfoundation/pyqrack.


**Table 1:** $F_c$ between two exact $3 \times 4$ chips. $w$: qubits in the widest simulator for $K = 3$.

| | $K=1$ | | $K=3$ | | |
|---|---|---|---|---|---|
| Interconnect | $d$=8 | $d$=12 | $d$=8 | $d$=12 | $w$ |
| ghost loom | 0.80 | 0.72 | 0.75 | 0.74 | 15 |
| mean-field loom | 0.69 | 0.59 | 0.49 | 0.49 | 12 |
| sample loom | 0.69 | 0.58 | 0.49 | 0.47 | 12 |
| links cut | 0.67 | 0.57 | 0.49 | 0.49 | 12 |
| ghost, stitched | 0.66 | 0.58 | 0.58 | 0.50 | 15 |
| bridge, $b = 1$ | 0.91 | 0.80 | 0.59 | 0.50 | 13 |
| bridge, $b = 2$ | 1.00 | 0.92 | 0.71 | 0.58 | 14 |
| bridge, $b = 3$ | 1.00 | 1.00 | 0.83 | 0.66 | 15 |
| bridge, $b = 4$ | 1.00 | 1.00 | 0.93 | 0.73 | 15 |

**Table 2:** $F_8$ inside a chip / across the link; two $3 \times 4$ chips, three links, depth 8.

| Inside | ghost | mean | cut |
|---|---|---|---|
| exact | .994/.985 | .964/.940 | .965/.941 |
| bridge, $b = 3$ | .970/.967 | .943/.911 | .946/.911 |
| bridge, $b = 2$ | .937/.943 | .912/.881 | .912/.881 |
| ACE 24:3:3 | .917/.931 | .888/.887 | .894/.885 |
| ACE 20:2:4 | .924/.948 | .887/.885 | .892/.888 |
| ACE 16:2:4 | .913/.927 | .886/.881 | .892/.888 |
| bridge chain, $b = 1, 2, 3, 4$: | .875/.832, .942/.933, .986/.986, .997/.998 | | |

**Table 3:** Nighthawk r3 on the ghost loom, six 8 GB dies.

| Links per pair | with ghost | skipped | peak GiB / chip |
|---|---|---|---|
| 4 | 28 | 0 | 0.5 to 1.0 |
| 6 | 42 | 0 | 1.0 to 2.7 |
| 8 | 56 | 4 | 1.6 to 3.8 |
| 10 | 64 | 8 | 3.0 to 4.4 |
| 10, ring | 57 | 3 | 2.4 to 3.5 |
| 8, hub | 36 | 1 | 0.3 to 5.3 |